Semi-Supervised Building Change Detection From Bitemporal Remote Sensing Images Leveraging Visual–Language Models and Consistency Learning

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***Abstract*—Building change detection (CD) from bitemporal remote sensing images aims to identify newly constructed, demolished, or modified buildings by comparing observations acquired at different times. A major limitation of existing supervised approaches is their strong dependence on pixelwise building and change annotations, which are costly and labor-intensive to obtain at scale. To address this challenge, we propose SemiBCD, a semi-supervised framework that integrates visual–language model (VLM) priors with consistency learning for building CD under limited supervision. A VLM pretrained on natural images is first employed to generate building pseudo-labels for remote sensing imagery. To mitigate domain shift and suppress noise in these pseudo-labels, we introduce an uncertainty-aware pseudo-label refinement (UAPLR) module that progressively improves pseudo-label reliability during training. Leveraging a small set of annotated change masks together with the refined pseudo-labels, we further design a dual consistency learning strategy, combining perturbation-based multiview consistency and temporal-hierarchical alignment to better exploit unlabeled data. Experiments on the LEVIR-CD and WHU-CD benchmarks, as well as the WHU Building dataset, demonstrate that SemiBCD consistently outperforms representative semi-supervised baselines under low-label regimes. The results indicate that incorporating VLM priors with uncertainty-aware refinement and consistency regularization provides an effective solution for reducing annotation dependence in building CD. The source code will be released upon publication at https://github.com/hejianglin2001/SemiBCD.git**

***Index Terms*—Building change detection (CD), consistency learning, semi-supervised learning, visual–language model (VLM).**

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I. Introduction

REMOTE sensing plays a pivotal role in environmental and urban monitoring [1], [2], [3]. Within this domain, change detection (CD) is a specific task within semantic CD [1], which aims to generate semantic segmentation (SS) maps for each temporal image and a binary change map indicating structural alterations. Supervised CD [1], [4], [5] typically relies on deep learning models trained with manual annotations. However, generating such annotations for bitemporal high-resolution imagery is labor-intensive, requires expert knowledge, and is often prohibitively expensive.

Recent semi-supervised building CD methods [6], [7], [8] have explored strategies such as consistency regularization or adversarial learning to exploit unlabeled data. While these approaches improve label efficiency, they remain fundamentally constrained by the limited semantic richness of the supervision signals derived from the CD network itself. In parallel, the emergence of large-scale visual–language models (VLMs), pretrained on massive image–text pairs, has opened new possibilities for injecting high-level semantic priors into remote sensing tasks. Several recent studies have begun to leverage VLMs to generate auxiliary semantic cues or pseudo-labels for CD, demonstrating encouraging performance gains.

However, existing VLM-assisted semi-supervised CD (SSCD) methods typically treat VLM outputs as *static* or uniformly reliable supervision. In practice, due to the significant domain gap between natural images and remote sensing imagery, VLM-generated pseudo-labels often exhibit spatially varying uncertainty, structural noise, and temporal inconsistency. Directly incorporating such noisy pseudo-labels into training can lead to suboptimal convergence and unstable learning, particularly under extremely limited labeled-data regimes, as illustrated in Fig. 1. Moreover, most prior works rely on a *single* form of consistency constraint, which is insufficient to jointly align semantic representations and temporal change patterns.

To address these limitations, we propose *SemiBCD*, a novel semi-supervised building CD framework integrating visual–language priors with consistency learning. Unlike existing approaches that simply combine VLM supervision with

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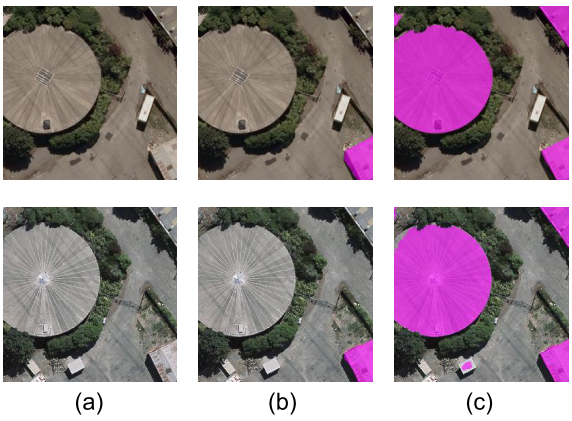


Fig. 1. Illustration of the need to refine building segmentation pseudo-labels generated by visual–language foundation models pretrained on natural images. The top and bottom rows correspond to two temporal phases (before and after the change). (a) Bitemporal remote sensing images: the first row shows the earlier time step, and the second row shows the later time step. (b) Initial building segmentation pseudo-labels produced by the VLM, exhibiting inaccuracies due to the domain shift between natural and remote sensing imageries. (c) Refined pseudo-labels generated using the proposed UAPLR module.

standard consistency regularization, our key insight is that *VLM-derived pseudo-labels must be explicitly refined and adaptively utilized*. To this end, we introduce an *uncertainty-aware pseudo-label refinement* (UAPLR) module that identifies and suppresses unreliable regions in VLM-generated building pseudo-labels before they are used for training. This refinement step is crucial for mitigating domain-shift-induced noise and stabilizing semi-supervised optimization.

Furthermore, SemiBCD employs a *dual consistency learning* strategy that enforces complementary constraints at different levels of representation. Specifically, *multiview consistency regularization* (MVCR) promotes prediction invariance under diverse data perturbations, while *temporal hierarchical alignment* (THA) explicitly aligns feature and semantic representations across temporal phases. Extensive ablation and sensitivity analyses demonstrate that these two components are mutually reinforcing and that neither alone can achieve comparable performance. In this way, SemiBCD goes beyond incremental integration and establishes a unified framework in which VLM priors, uncertainty modeling, and multilevel consistency are jointly optimized.

Extensive experiments on the LEVIR-CD [9] and WHU-CD [10] datasets under multiple labeled-data ratios validate the effectiveness of the proposed approach. SemiBCD consistently outperforms existing semi-supervised and VLM-assisted baselines, particularly in low-label regimes, while maintaining competitive inference efficiency. These results confirm that explicitly modeling pseudo-label uncertainty and enforcing dual consistency constitute a meaningful methodological advance for semi-supervised building CD.

II. Related Work

A. Supervised CD

With the rapid development of deep learning, supervised CD methods have achieved remarkable performance by leveraging convolutional and transformer-based architectures trained with manual annotations. Depending on output type, existing supervised methods can be broadly categorized into binary CD and semantic CD [1].

1) Binary CD: Binary CD focuses on distinguishing changed and unchanged regions. Early deep learning-based approaches typically adopted Siamese convolutional networks to extract multitemporal features. For example, Chen et al. [11] directly fused multilevel features from two temporal streams, while hierarchical fusion networks [12], [13] aggregated features at multiple scales to capture both global context and fine spatial details. These architectures improved the detection of subtle or small-scale changes but often suffered from limited receptive fields and redundant feature computation.

To address such issues, transformer-based models have recently been introduced to CD for their ability to capture long-range dependencies. Bitemporal image transformer (BiT) [14] utilized a transformer encoder to establish spatial–temporal correspondences, while progressive sampling transformer network [15] refined change localization through progressive spatial–temporal optimization. Changer [16] proposed feature interaction between temporal branches, and attention-based multiscale transformer network (AMTNet) [17] integrated convolutional neural networks (CNNs) and transformers for multiscale feature aggregation and contextual alignment. Similarly, multiscale context-aware network [18] exploited hybrid attention to enhance temporal consistency. Despite the success of these transformer-based models, they typically require large annotated datasets to achieve stable performance—an unrealistic assumption in remote sensing, where collecting change labels is time-consuming and labor-intensive.

2) Semantic CD: Semantic CD extends binary CD by predicting SS maps for both timestamps, along with a change mask indicating category transitions. This task provides richer scene-level understanding, such as differentiating new construction from vegetation loss.

Earlier CNN-based semantic CD frameworks [4], [20]—e.g., dual-task constrained deep Siamese convolutional network [20]—jointly learned land-cover segmentation and change maps using multitask or attention-based structures. Xia et al. [21] adopted deep Siamese networks to fuse postclassification results, while symmetric multitask network [22] introduced multicontent fusion modules to enhance feature discrimination. Although these models captured complex local variations, they remained limited by the restricted receptive field and insufficient cross-temporal interaction of CNNs.

With the advent of transformers, several works have explored global spatial–temporal modeling for semantic CD. ChangeMask [5] utilized an encoder–decoder transformer for contextual fusion, and multitask learning for semantic CD network [23] adopted shared encoders with task-specific decoders to improve semantic consistency. Liu et al. [1] proposed a multitask CNN-transformer network (MCTNet) integrating CNN and transformer modules for hierarchical feature extraction, enforcing temporal symmetry for more stable detection. Despite these efforts, supervised semantic CD

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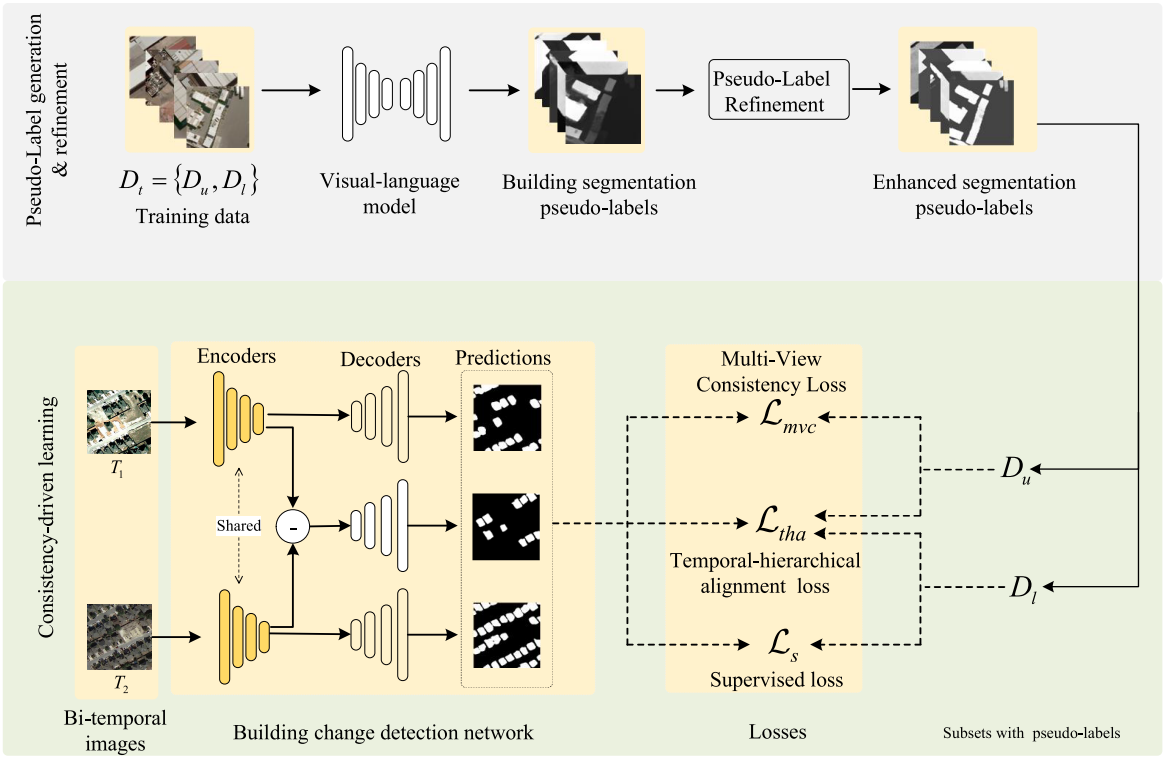


Fig. 2. Overview of the proposed framework. Given a bitemporal remote sensing image pair, our SemiBCD outputs building segmentation maps for both timestamps and a change mask indicating building modifications. It leverages VLMs for pseudo-label generation and incorporates consistency learning to achieve high-accuracy CD with minimal labeled data. The training process includes two stages: 1) generating initial building segmentation pseudo-labels with a pretrained VLM (APE [19]), followed by self-training to progressively refine them and 2) applying a dual consistency learning strategy that enforces prediction consistency under perturbations, thereby enhancing robustness and detection accuracy.

methods remain heavily annotation-dependent and often fail to generalize across domains or geographical scenes due to limited labeled-data diversity.

B. Semi-Supervised CD

SSCD addresses the challenge of limited labeled data by leveraging both labeled and unlabeled samples to improve detection accuracy. Early methods, such as [24], introduced a coarse-to-fine SSCD approach for multispectral images, while Peng et al. [6] proposed SemiCDNet, a CNN utilizing unlabeled data for high-resolution CD. Recent advances have incorporated contrastive learning [25], graph attention networks [26], and consistency regularization [27] to enhance model generalization and robustness.

Self-training has emerged as a key technique, with methods such as STCRNet [28] integrating self-training and consistency regularization, and Mao et al. [29] employing a mean teacher model for pseudo-label construction. Advanced strategies, including pseudo-data generation [8], [30], [31], adversarial learning [32], and domain adaptation [33], [34], [35], further improve performance across diverse remote sensing scenarios.

Transformer-based SSCD approaches [36], [37] have gained attention for their ability to model long-range dependencies, while ensemble learning and cross-supervision methods [38], [39] refine pseudo-label reliability. Prototype-based learning [40] and uncertainty modeling [41], [42] offer additional advancements in reducing semantic inconsistencies. Recent studies emphasize perturbation-based consistency learning [43] and adaptive pseudo-label evaluation [42] to further boost SSCD performance.

Despite recent progress, most SSCD methods treat pseudo-label generation and consistency regularization as separate components and often assume that basic filtering yields sufficiently reliable pseudo-labels. However, pseudo-labels from pretrained vision–language models can propagate errors due to domain shifts between natural-image pretraining and remote sensing data. Explicitly modeling pseudo-label uncertainty while jointly optimizing refinement and consistency remains underexplored in building CD, which motivates our proposed framework.

III. Proposed Framework

A. Overview of the Proposed Framework

In the semi-supervised training setting, we utilize a small labeled dataset, $\mathcal{D}_l = \{\mathbf{x}_n^l, \mathbf{y}_n^l\}_{n=1}^{N_l}$ (with change labels $\mathbf{y}_n^l$), and a much larger unlabeled dataset, $\mathcal{D}_u = \{\mathbf{x}_n^u\}_{n=1}^{N_u}$, where $N_u \gg N_l$. Each sample $\mathbf{x}_n^* = (\mathbf{I}_{n,1}^*, \mathbf{I}_{n,2}^*)$ (with $* \in \{l, u\}$) represents a bitemporal image pair of pre-event ($\mathbf{I}_{n,1}^*$) and postevent ($\mathbf{I}_{n,2}^*$) images. The total training dataset is $\mathcal{D}_t = \mathcal{D}_l \cup \mathcal{D}_u$.

We propose SemiBCD, a semi-supervised framework that combines VLMs and consistency learning. As illustrated in Fig. 2, SemiBCD operates in two stages.

1) *Pseudo-Labels Generation and Refinement:* A pretrained VLM produces initial building segmentation masks for

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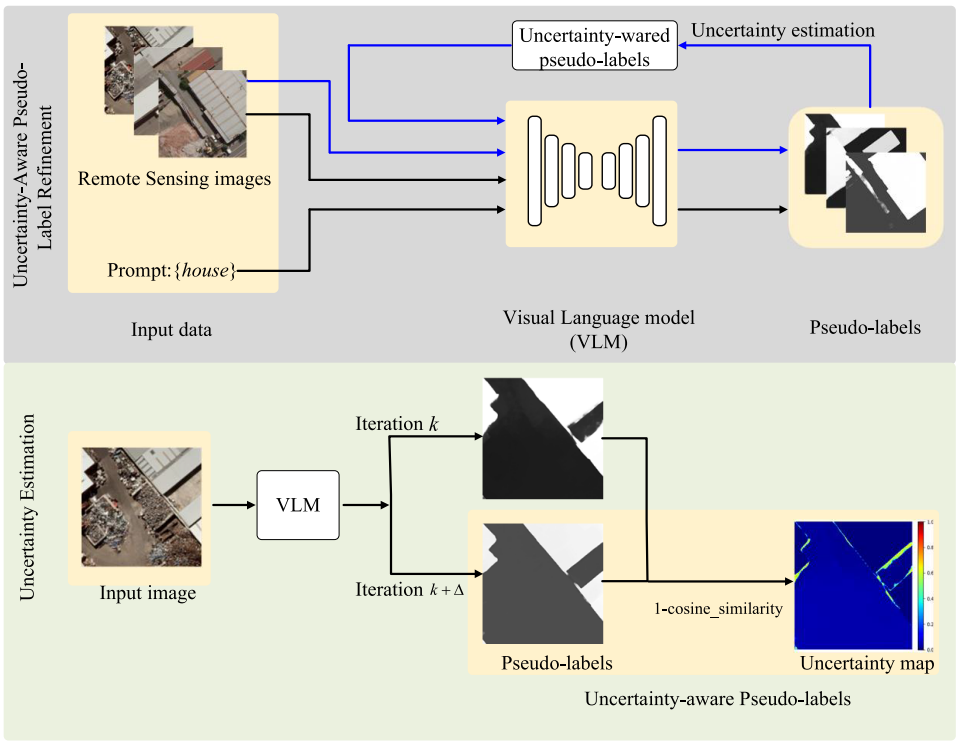


Fig. 3. Illustration of pseudo-label generation and refinement for building segmentation with VLMs. In the subfigure, the black arrowed lines correspond to the process of generating initial pseudo-labels, while the blue arrowed lines correspond to the retraining process using the pseudo-labels.

all samples in $\mathcal{D}_t$. These masks are progressively refined through uncertainty-aware self-training to enhance label quality without manual annotation. Change masks for $\mathcal{D}_u$ are then derived by differencing the refined building masks.

2) *Consistency-Driven Training:* The building CD model learns from the training data in $\mathcal{D}_t$ using both the refined pseudo-labels and the limited manually labeled change labels through perturbation-based *MVCR* and *THA* consistency learning.

The primary aim of this dual-stage approach is to significantly reduce reliance on extensive manual labeling while maintaining high detection accuracy by harnessing the complementary strengths of VLMs (zero-shot generalization) and consistency learning (robust feature representation).

B. Pseudo-Label Generation and Refinement for SS With VLMs

Due to the inherent domain gap between natural and remote sensing imagery, directly applying pretrained VLMs can introduce noise and inaccuracies. To ensure high-quality pseudo-labels are generated, our proposed pseudo-label generation framework comprises two stages, as illustrated in Fig. 3: 1) initial building mask generation using VLMs, followed by 2) UAPLR.

1) Initial Building Mask Generation: Without loss of generality, we employ an aligned prompt encoder (APE) [19], a state-of-the-art (SOTA) VLM, to generate building masks by aligning textual prompts with visual features in bitemporal imagery. Given a bitemporal image pair $(I_{n,1}, I_{n,2})$ from the training set $\mathcal{D}_t$, we utilize APE to produce initial building masks through text-driven semantic alignment. The text encoder is prompted with the query *house*. APE's visual encoder then processes $I_{n,t}(t \in \{1,2\})$ to extract multiscale features fused with the text embeddings via cross-modal attention. This interaction generates a set of nonoverlapping building instances, which are then merged into the initial building mask $P_{n,t} \in \mathbb{R}^{H\times W}$. $P_{n,t}$ is a pixelwise probability map, where each value $P_{n,t}(i,j) \in [0,1]$ reflects the likelihood of pixel (i,j) in image $I_{n,t}(t \in \{1,2\})$ belonging to a building.

Due to domain shifts between APE's pretraining data (natural images) and remote sensing imagery, the model's confidence scores for buildings are often conservatively calibrated. To address the domain shifts, we adopt a permissive threshold $\tau_{\text{low}}(e.g., 0.25)$ to binarize the probability maps

$$\hat{M}^s_{n,t}(i,j) = \begin{cases} 1, & P_{n,t}(i,j) \geq \tau_{\text{low}} \\ 0, & \text{otherwise} \end{cases} \quad t \in \{1,2\}. \tag{1}$$

This lower threshold balances recall and precision, retaining regions with moderate building signatures (e.g., partial structures or spectrally ambiguous rooftops) while accepting controlled noise. The resulting masks $\hat{M}^s_{n,1}$ and $\hat{M}^s_{n,2}$ serve as the initial building segmentation pseudo-labels, which are subsequently refined through uncertainty quantification to filter spurious activations.

2) UAPLR: As shown in Fig. 4, the pseudo-labels produced by APE are unstable during the fine-tuning process when using these initial masks. Applying a fixed threshold for noise filtering can be overly simplistic and may either discard valuable information or retain misleading labels—particularly

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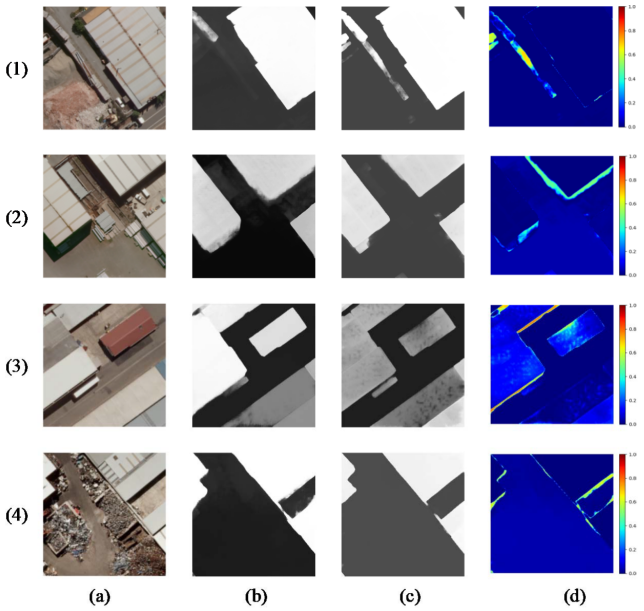


Fig. 4. Illustration of prediction uncertainty: (a) input image; (b) and (c) SoftMax prediction outcomes generated by APE trained for 3000 and 6000 iterations, respectively, with grayscale images representing prediction values ranging from 0 to 1; and (d) discrepancies between the predictions in (b) and (c).

under domain shift conditions. Furthermore, as illustrated in Fig. 4(b) and (c), noisy regions are predominantly concentrated around building edges, which are particularly susceptible to prediction fluctuations during training. Motivated by this observation, we propose the UAPLR module to refine the building segmentation masks, as presented in Fig. 3.

Our UAPLR module dynamically assesses prediction stability through cosine similarity. It allows for a more nuanced handling of uncertainty, where stable and reliable predictions are emphasized while noisy or inconsistent ones are de-emphasized. Specifically, for each pixel (i, j) in image $I_{n,t}$, we compare the building mask predictions before and after a certain number of training iterations to assess prediction stability. Let $P_{n,t}^{(k)}(i, j)$ and $P_{n,t}^{(k+\Delta)}(i, j)$ denote the predicted probabilities for pixel (i, j) at iterations k and $k + \Delta$, respectively, where Δ represents the interval between evaluations. We then compute the cosine similarity $S_{n,t}(i, j)$ between these predictions as a measure of consistency

$$S_{n,t}(i,j) = \frac{P_{n,t}^{(k)}(i,j) \cdot P_{n,t}^{(k+\Delta)}(i,j)}{\|P_{n,t}^{(k)}(i,j)\| \cdot \|P_{n,t}^{(k+\Delta)}(i,j)\|}. \tag{2}$$

Pixels with high cosine similarity ($S_{n,t}(i, j) \approx 1$) indicate stable predictions across iterations and are assigned greater weight in the loss function, while inconsistent predictions ($S_t(i, j) \ll 1$) are downweighted to reduce their influence, thereby refining the quality of the pseudo-labels and improving overall model performance.

Fig. 4(d) visualizes the discrepancies between the predictions in Fig. 4(b) and (c), using the cosine similarity metric defined in (2). As shown, regions with low similarity values—indicating high prediction inconsistency—are predominantly located along building edges. By incorporating this similarity metric into the loss function as a weighting factor, as described in (2), our method shows promise in suppressing the impact of noisy predictions during training.

Computed using the standard cross-entropy loss, the weighted segmentation loss $\mathcal{L}_{\text{ws}}$ employed to refine the building masks in image $I_{n,t}$ can be formulated as

$$\mathcal{L}_{ws}(I_{n,t}) = -\frac{1}{H \times W} \sum_{(i,j)\in\Omega} w_t(i,j) \left[\hat{M}_{n,t}^{s}(i,j) \log P_{n,t}^{(k+\Delta)}(i,j) + \left(1 - \hat{M}_{n,t}^{s}(i,j)\right) \log\left(1 - P_{n,t}^{(k+\Delta)}(i,j)\right)\right]. \tag{3}$$

Here, Ω denotes the spatial domain of the image with height H and width W. $w_t(i, j) = S_t(i, j)$ is the cosine similarity weights. By emphasizing pixels with consistent predictions, the loss function guides APE to focus on reliable regions, effectively reducing noise in the building masks.

The training objective to fine-tune APE is a linear combination of classification loss, localization loss, and the weighted segmentation loss for the encoder and decoder, respectively. It can be formulated as

$$\mathcal{L}_p(\mathcal{D}_l, \mathcal{D}_u) = \mathcal{L}_{cls}(\mathcal{D}_l, \mathcal{D}_u) + \mathcal{L}_{dice}(\mathcal{D}_l, \mathcal{D}_u) + \mathcal{L}_{ws}(\mathcal{D}_l, \mathcal{D}_u) \tag{4}$$

where $\mathcal{L}_{cls}$ is the focal loss [44] to classify building and background regions for the encoder and align language and vision embeddings for the decoder. $\mathcal{L}_{dice}$ is the dice loss [45], which supervises the last output of the decoder only.

For each unlabeled sample $\mathbf{x}_n^u = (\mathbf{I}_{n,1}^u, \mathbf{I}_{n,2}^u)$ in the training data $\mathcal{D}_u$, we generate two refined building segmentation masks, $\hat{M}_{n,1}^{u,s}$ for the pre-event image $\mathbf{I}_{n,1}^u$ and $\hat{M}_{n,2}^{u,s}$ for the postevent image $\mathbf{I}_{n,2}^u$), respectively. The pseudo change mask $\hat{\mathbf{y}}_n^{c,u}$ is then obtained by pixelwise comparison of $\hat{M}_{n,1}^{u,s}$ and $\hat{M}_{n,2}^{u,s}$

$$\hat{\mathbf{y}}_n^{c,u}(i,j) = \begin{cases} 0, & \hat{M}_{n,1}^{u,s}(i,j) = \hat{M}_{n,2}^{u,s}(i,j) \\ 1, & \text{otherwise.} \end{cases} \tag{5}$$

C. Building CD Based on Consistency Learning

We introduce two strategies to enable the building CD model to learn effectively from unlabeled data and facilitate mutual enhancement between the two subtasks during training. The first strategy is perturbation-based *MVCR*, which leverages unlabeled data to improve generalization. The second is *THA*, designed to ensure temporal coherence at both the feature representation and semantic prediction levels by incorporating change mask information.

1) Supervised Loss: For each labeled bitemporal pair $\mathbf{x}_n^l = (\mathbf{I}_{n,1}^l, \mathbf{I}_{n,2}^l)$ in the labeled training set $\mathcal{D}_l = \{\mathbf{x}_n^l, \mathbf{y}_n^{c,l}\}_{n=1}^{N_l}$, we have a manually labeled change mask $\mathbf{y}_n^{c,l}$ along with two refined building segmentation masks, $\hat{M}_{n,1}^{l,s}$ and $\hat{M}_{n,2}^{l,s}$. For the labeled data from $\mathcal{D}_l$, the supervised loss integrates the objectives of building segmentation and CD

$$\mathcal{L}_s(\mathcal{D}_l) = \mathcal{L}_{\text{ss}}(\mathcal{D}_l) + \mathcal{L}_{\text{cd}}(\mathcal{D}_l) \tag{6}$$

where $\mathcal{L}_{\text{ss}}$ optimizes building segmentation accuracy, and $\mathcal{L}_{\text{cd}}$ refines CD performance. Both $\mathcal{L}_{\text{ss}}$ and $\mathcal{L}_{\text{cd}}$ are computed using the standard cross-entropy loss.

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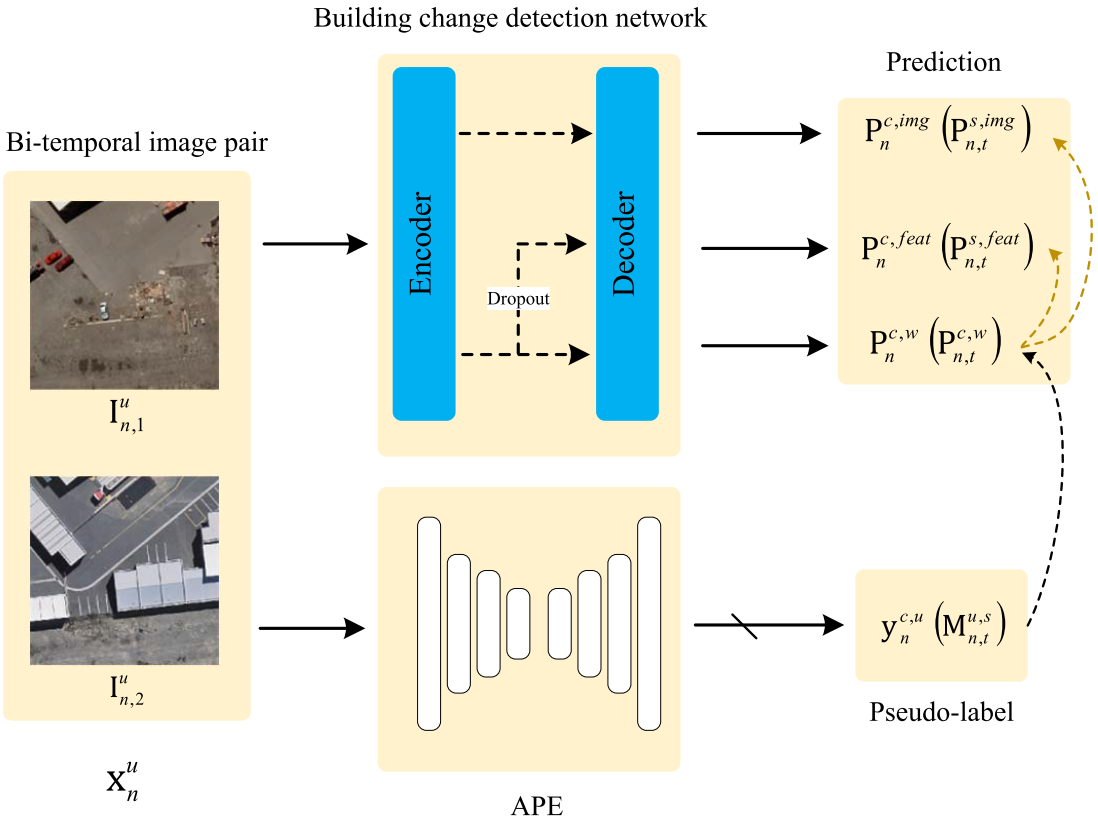


Fig. 5. Illustration of the multiview consistency learning with unlabeled training data.

For each labeled sample $\mathbf{x}_n^l$, the segmentation loss $\mathcal{L}_{\text{ss}}$ between the predicted building mask $P_{n,t}^s$ and the corresponding pseudo-labels $\hat{M}_{n,t}^{l,s}$ can be mathematically expressed as

$$\begin{aligned}\mathcal{L}_{\text{ss}}\left(\mathcal{D}_l\right) &= \frac{1}{2N_l}\sum_{n=1}^{N_l}\sum_{t\in\{1,2\}}\mathcal{L}_{ce}\left(\hat{M}_{n,t}^{l,s}, P_{n,t}^s\right)\\ &= -\frac{1}{2N_lHW}\sum_{n=1}^{N_l}\sum_{t\in\{1,2\}}\sum_{(i,j)\in\Omega}\hat{M}_{n,t}^{l,s}(i,j)\log P_{n,t}^s(i,j)\end{aligned} \tag{7}$$

where $P_{n,t}^s(i,j)$ represents the predicted probability of pixel (i,j) belonging to the building class at timestamp t.

The CD loss $\mathcal{L}_{\text{cd}}$ between the predicted change mask $P_n^c(i,j)$ and the manually labeled change mask $\mathbf{y}_n^{c,l}$ can be formulated as

$$\begin{aligned}\mathcal{L}_{\text{cd}}\left(\mathcal{D}_l\right) &= -\frac{1}{N_l}\sum_{n=1}^{N_l}\mathcal{L}_{ce}\left(\mathbf{y}_n^{c,l}, P_n^c\right)\\ &= -\frac{1}{N_lHW}\sum_{n=1}^{N_l}\sum_{(i,j)\in\Omega}\mathbf{y}_n^{c,l}(i,j)\log P_n^c(i,j).\end{aligned} \tag{8}$$

Here, $P_n^c(i,j)$ represents the predicted probability of change at pixel (i,j).

2) MVCR Loss: To bolster model robustness and efficiently exploit unlabeled data, we introduce a multiview consistency loss, as illustrated in Fig. 5, which mandates prediction consistency across varying perturbation intensities. For an unlabeled sample $\mathbf{x}_n^u = (\mathbf{I}_{n,1}^u, \mathbf{I}_{n,2}^u)$, we apply both weak (minor geometric and color augmentations) and strong (image-level and feature-level transformations) perturbations. This mechanism encourages the model to make stable predictions, which are enforced through three specific consistency losses, detailed in the following.

The pseudo-to-weak consistency loss $\mathcal{L}_{\text{p2w}}$ ensures consistency between the model's predictions under weak perturbations and the corresponding pseudo-labels. The loss function is formulated as follows:

$$\begin{aligned}\mathcal{L}_{\text{p2w}}\left(\mathcal{D}_u\right) &= \frac{1}{2N_u}\sum_{n=1}^{N_u}\sum_{t\in\{1,2\}}\mathcal{L}_{ce}\left(\hat{M}_{n,t}^{u,s}, P_{n,t}^{s,w}\right)\\ &\quad+\frac{1}{N_u}\sum_{n=1}^{N_u}\mathcal{L}_{ce}\left(\hat{\mathbf{y}}_n^{c,u}, P_n^{c,w}\right)\end{aligned} \tag{9}$$

where $P_{n,t}^{s,w}$ and $P_n^{c,w}$ represent the predicted segmentation and CD outputs under weak perturbation, respectively. $\hat{M}_{n,t}^{u,s}$ and $\hat{\mathbf{y}}_n^{c,u}$ serve as pseudo-labels for SS and CD.

We employ weak-to-image and weak-to-feature consistency losses to enforce consistency between weakly perturbed and strongly perturbed predictions, ensuring that the model's outputs remain aligned across different levels of data augmentation, from minor transformations to significant distortions. These functions can be expressed as

$$\begin{aligned}\mathcal{L}_{\text{w2i}}\left(\mathcal{D}_u\right) &= \frac{1}{2N_u}\sum_{n=1}^{N_u}\sum_{t\in\{1,2\}}\mathcal{L}_{ce}\left(P_{n,t}^{s,w}, P_{n,t}^{s,\text{img}}\right)\\ &\quad+\frac{1}{N_u}\sum_{n=1}^{N_u}\mathcal{L}_{ce}\left(P_n^{c,w}, P_n^{c,\text{img}}\right)\end{aligned} \tag{10}$$

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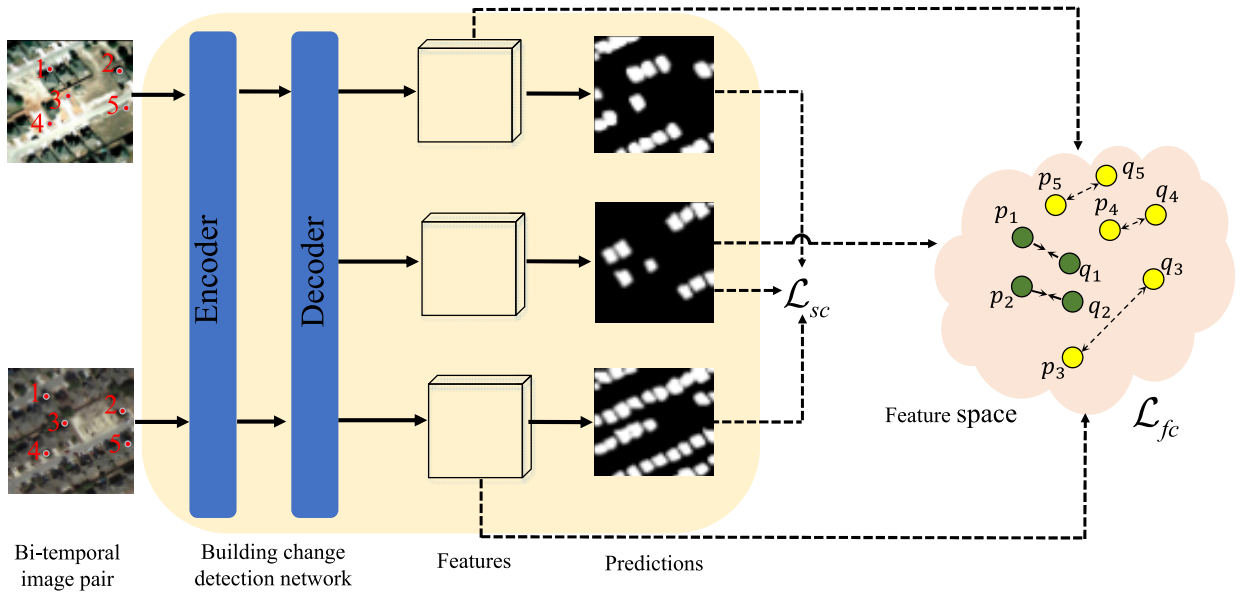


Fig. 6. Illustration of the THA loss. The yellow dots correspond to pixel pairs with predicted changes, while the green dots indicate unchanged pixel pairs.

$$\mathcal{L}_{\mathrm{w2f}}\left(\mathcal{D}_u\right)=\frac{1}{2N_u}\sum_{n=1}^{N_u}\sum_{t\in\{1,2\}}\mathcal{L}_{ce}\left(P_{n,t}^{s,w},P_{n,t}^{s,\mathrm{feat}}\right)+\frac{1}{N_u}\sum_{n=1}^{N_u}\mathcal{L}_{ce}\left(P_{n}^{c,w},P_{n}^{c,\mathrm{feat}}\right). \tag{11}$$

By integrating all consistency losses, the final MVCR loss is given by

$$\mathcal{L}_{\mathrm{mvc}}\left(\mathcal{D}_u\right)=\mathcal{L}_{\mathrm{p2w}}\left(\mathcal{D}_u\right)+\mathcal{L}_{\mathrm{w2i}}\left(\mathcal{D}_u\right)+\mathcal{L}_{\mathrm{w2f}}\left(\mathcal{D}_u\right). \tag{12}$$

3) THA Loss: Our framework incorporates the THA module to maintain temporal coherence at both feature and semantic levels, as shown in Fig. 6. This design stabilizes unchanged regions while enhancing discriminative cues for meaningful changes. The synergy between segmentation and CD enables mutual reinforcement.

For $\mathbf{x}_n=(\mathbf{I}_{n,1},\mathbf{I}_{n,2})\in\mathcal{D}_t$, let $d_n(i,j)=\|F_{n,1}^s(i,j)-F_{n,2}^s(i,j)\|_2^2$ represent the squared L_2 distance between pixel embeddings at (i,j), where $F_{n,1}^s(i,j)$ and $F_{n,2}^s(i,j)$ are the pixel embeddings from segmentation decoders. The feature-level consistency loss for x_n at (i,j) is then defined as

$$\mathcal{L}_{fc}\left(x_n(i,j)\right)=\begin{cases}d_n(i,j), & y_n^c(i,j)=0\\ \max\left(0,m-d_n(i,j)\right), & y_n^c(i,j)=1\end{cases} \tag{13}$$

where $y_n^c(i,j)$ represents the predicted or the manually labeled label at (i,j). m is a margin hyperparameter that enforces a minimum separation between changed features, empirically set to 1.0.

Let $P_{n,1}^s(i,j)$ and $P_{n,2}^s(i,j)$ denote the predicted building segmentation probabilities for $I_{n,1}$ and $I_{n,2}$ at pixel (i,j), respectively. The semantic-level consistency loss for x_n at pixel (i,j) is expressed as

$$\mathcal{L}_{sc}\left(x_n(i,j)\right)=\begin{cases}\cos\left(P_{n,1}^s(i,j),P_{n,2}^s(i,j)\right), & y_n^c(i,j)=0\\ 1-\cos\left(P_{n,1}^s(i,j),P_{n,2}^s(i,j)\right), & y_n^c(i,j)=1.\end{cases} \tag{14}$$

TABLE I

CORRESPONDENCE BETWEEN THE THREE LOSS COMPONENTS ($\mathcal{L}s$, $\mathcal{L}mvc$, AND $\mathcal{L}_{tha}$) AND THEIR ASSOCIATED TRAINING CONFIGURATIONS FOR SS AND CD TASKS

Loss	Task	Subset usage		Label		Perturbations	
		$\mathcal{D}_l$	$\mathcal{D}_u$	$\mathcal{D}_l$	$\mathcal{D}_u$	Weak	Strong
$\mathcal{L}_s$	SS	✓	×	PL	-	×	×
	CD	✓	×	✓	-	×	×
$\mathcal{L}_{mvc}$	SS	×	✓	-	PL	✓	✓
	CD	×	✓	-	PL	✓	✓
$\mathcal{L}_{tha}$	SS	✓	✓	PL	PL	×	×
	CD	✓	✓	✓	PL	×	×

PL represents pseudo-labels. The symbols ✓ and × denote the presence or absence of the corresponding items, respectively. - represents not applicable.

This formulation enforces semantic similarity in unchanged regions while encouraging separation in changed areas, reinforcing the hierarchical consistency framework.

The combined THA loss $\mathcal{L}_{tha}$ for the training set $\mathcal{D}_t$ is formulated as

$$\mathcal{L}_{tha}\left(\mathcal{D}_l,\mathcal{D}_u\right)=\frac{1}{|\mathcal{D}_t|}\sum_{\mathbf{x}_n\in\mathcal{D}_t}\left(\mathcal{L}_{fc}\left(x_n\right)+\mathcal{L}_{sc}\left(x_n\right)\right). \tag{15}$$

4) Total Loss: The total loss consists of three components: the supervised loss $\mathcal{L}_s(\mathcal{D}_l)$, the MVCR loss $\mathcal{L}_{mvc}(\mathcal{D}_u)$, and the THA loss $\mathcal{L}_{tha}(\mathcal{D}_l,\mathcal{D}_u)$. Correspondence between the three loss components and their associated training configurations for SS and CD is summarized in Table I. The total loss of our semi-supervised building CD framework can be expressed as

$$\mathcal{L}_{\mathrm{total}}=\mathcal{L}_s\left(\mathcal{D}_l\right)+\lambda_1\mathcal{L}_{mvc}\left(\mathcal{D}_u\right)+\lambda_2\mathcal{L}_{tha}\left(\mathcal{D}_l,\mathcal{D}_u\right). \tag{16}$$

The weighting factors λ_1 and λ_2 balance the contributions of the semi-supervised consistency loss and the THA loss. The total loss jointly optimizes the SS and CD tasks.

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TABLE II

COMPARISONS OF SEMIBCD WITH OTHER SEMI-SUPERVISED METHODS ON LEVIR-CD AND WHU-CD TESTING SETS (%) ↑

Method	Backbone	LEVIR-CD (% labeled) → LEVIR-CD				WHU-CD (% labeled)→ WHU-CD			
		5%(356) IoU^c / OA	10%(712) IoU^c / OA	20%(1,424) IoU^c / OA	40%(2,848) IoU^c / OA	5%(297) IoU^c / OA	10%(594) IoU^c / OA	20%(1,189) IoU^c / OA	40%(2,378) IoU^c / OA
Super.only *	ResNet50	61.0 / 97.60	66.8 / 98.13	72.3 / 98.44	74.9 / 98.60	50.0 / 97.48	55.7 / 97.53	65.4 / 98.20	76.1 / 98.94
AdvNet [48]*	ResNet101	66.1 / 98.08	72.3 / 98.45	74.6 / 98.58	75.0 / 98.60	55.1 / 97.90	61.6 / 98.11	73.8 / 98.80	76.6 / 98.94
s4GAN [49]*	ResNet101	64.0 / 97.89	67.0 / 98.11	73.4 / 98.51	75.4 / 98.62	- / 96.69	62.6 / 98.15	70.8 / 98.60	76.4 / 98.96
FixMatch [50]	ResNet50	79.96 / 98.92	81.66 / 99.01	81.83 / 99.02	82.12 / 99.04	75.72 / 98.92	80.29 / 99.14	81.27 / 99.17	84.38 / 99.33
UniMatch [51]	ResNet50	80.12 / 98.92	81.70 / 99.02	81.72 / 99.01	81.56 / 99.00	80.17 / 99.15	81.44 / 99.21	81.72 / 99.19	85.86 / 99.39
SemiCDNet [6]*	UNet++	67.6 / 98.17	71.5 / 98.42	74.3 / 98.58	75.5 / 98.63	51.7 / 97.71	62.0 / 98.16	66.7 / 98.28	75.9 / 98.93
SemiCD [7]*	ResNet50	72.5 / 98.47	75.5 / 98.63	76.2 / 98.68	77.2 / 98.72	65.8 / 98.37	68.1 / 98.47	74.8 / 98.84	77.2 / 98.96
SemiCD-VL [46]	ResNet50	81.83 / 98.99	82.51 / 99.05	82.63 / 99.05	82.84 / 99.07	81.65 / 99.22	82.98 / 99.28	84.12 / 99.32	85.66 / 99.39
SemiBCD (ours)	ResNet50	83.17 / 99.08	83.75 / 99.12	83.92 / 99.13	84.24 / 99.14	83.34 / 99.32	84.32 / 99.35	87.08 / 99.45	88.50 / 99.52

Results marked with * are from [7]. The value "18.3" was replaced with "–" due to unreliability. UniMatch and SemiCD-VL were reproduced using their official codebases; FixMatch was implemented within the UniMatch framework for fair comparison, as it supports general semi-supervised segmentation and has been extended to change detection in prior work. Ranking colors: red = 1st, blue = 2nd, green = 3rd best in IoU^c and OA per column.

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: Following prior works on building CD [7], [46], we evaluated our method on two widely used benchmarks: *LEVIR-CD* [9] and *WHU-CD* [10]. *LEVIR-CD* consists of 637 pairs of very high-resolution (0.5 m/pixel) Google Earth images (1024 × 1.024 pixels), captured over 5–14 years to reflect urban expansion and demolition. It includes 31 333 annotated building change instances across diverse urban and suburban areas, with expert-labeled pixel-level masks. *WHU-CD* contains ultrahigh-resolution (0.075 m/pixel) aerial images of Christchurch, New Zealand, covering 450 km^2 and focusing on building changes caused by the 2011 earthquake. It provides pixel-level annotations for over 12 000 and 16 000 buildings in 2012 and 2016 imageries, respectively. For both datasets, only pixel-level change masks were provided as ground truth. Images were divided into nonoverlapping 256 × 256 patches.

Specifically, LEVIR-CD and WHU-CD were split into 7120/1024/2048 and 5947/743/744 patches for training, validation, and testing, respectively. In the semi-supervised setting, a subset of the training data $\mathcal{D}_t$ was labeled ($\mathcal{D}_l$), and the remainder was treated as unlabeled ($\mathcal{D}_u$) by discarding change masks. SemiBCD was evaluated under four labeled-data ratios: 5%, 10%, 20%, and 40%.

Additionally, we employed the widely used *WHU Building dataset* [10] (downloaded from Hugging Face) to quantitatively assess the effectiveness of our UAPLR module for building segmentation. This dataset contains 512 × 512 pixel images and is split into 4736, 1036, and 500 samples for training, validation, and testing, respectively.

2) Metrics: To evaluate the performance of building CD, we adopted three widely used metrics [1], [17]: intersection over union (IoU) for the change class (IoU^c), $F1$-score for the change class (F_1^c), and overall accuracy (OA). For building segmentation, we reported four standard metrics: IoU for the building class (IoU), $F1$-score, precision, and recall.

3) Implementation Details: The framework was implemented in PyTorch and trained on an NVIDIA RTX 3090 GPU (24-GB memory). We used the Adam optimizer [47] with default settings, except for the following: momentum = 0.9, weight decay = $1e^{-}4$, and an initial learning rate of $1e^{-}3$, which was reduced by a factor of 10 every 20 epochs. Training spanned 100 epochs with a batch size of 8, requiring approximately 150 total GPU hours (about 1.5 h/epoch). The consistency regularization weights are set to $\lambda_1 = \lambda_2 = 0.3$, and the uncertainty threshold τ_{low} is fixed at 0.2. During pseudo-label refinement, the cosine similarity between building mask predictions is computed every $\Delta = 3000$ iterations. All hyperparameters in our framework were selected using the validation set of each dataset.

B. Comparison Results Within the Same Dataset

Following previous research [7], [46], we first evaluated our SemiBCD under the *within-dataset setting*, where both training and testing were performed on the same dataset. We compared our approach against representative SOTA semi-supervised learning baselines, including general-purpose methods—adversarial network (AdvNet) [48], semi-supervised segmentation via generative adversarial networks (s4GAN) [49], FixMatch [50], and unified dual-stream perturbations (UniMatch) [51]—as well as CD-specific methods: SemiCD-Net [6], SSCD [7], and visual–language-based SemiCD (SemiCD-VL) [46]. We also included a fully supervised baseline, referred to as Super.only, which was trained exclusively on the labeled samples. Following UniMatch and SemiCD-VL, we implemented our semi-supervised learning scheme using the DeepLabv3+ [53] SS architecture with a ResNet-50 [54] backbone. For reproducibility and fairness, both our method and all reimplemented baselines in Table II were evaluated over three independent runs, with results reported as averages.

As shown in Table II, all semi-supervised methods markedly outperform the Super.only baseline, highlighting the value of unlabeled data. SemiBCD achieves the best results across all label ratios, with especially large gains under limited supervision. For instance, with only 5% labels on LEVIR-CD, it attains an IoU^c of 83.17%, surpassing UniMatch (80.12%) and SemiCD-VL (81.83%) by 3.05% and 1.34%, respectively. On WHU-CD, SemiBCD further improves IoU^c by 1.69% over SemiCD-VL, confirming its ability to generate and exploit reliable pseudo-labels.

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TABLE III

PERFORMANCE COMPARISON WITH STATISTICAL SIGNIFICANCE ANALYSIS ON THE LEVIR-CD AND WHU-CD DATASETS IN TERMS OF CHANGE-CLASS IOU (IoU^c) USING ONLY 5% LABELED TRAINING DATA. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER THREE RUNS

Method	LEVIR-CD	WHU-CD
FixMatch [52]	79.96 ± 0.56	75.72 ± 0.57
UniMatch [14]	80.12 ± 0.53	80.17 ± 0.56
SemiCD-VL [46]	81.83 ± 0.43	81.65 ± 0.36
SemiBCD (Ours)	**83.17** ± **0.11**	**83.34** ± **0.21**

When more labeled data are available, the performance of all models improves, yet SemiBCD maintains a consistent advantage. With 20% labels on LEVIR-CD, SemiBCD achieves 83.92% IoU^c, surpassing SemiCD-VL (82.63%) and UniMatch (81.72%). On WHU-CD with 40% labels, SemiBCD further pushes IoU^c to 88.50%, outperforming SemiCD-VL (85.66%) by 2.84%. Notably, general-purpose semi-supervised methods such as FixMatch and UniMatch tend to underperform in complex bitemporal scenarios because they do not explicitly model spatial correspondences or structural regularities of buildings. In contrast, SemiBCD exploits refined building segmentation masks to guide pseudo change labeling, reducing the noise often present in automatically generated supervision and yielding more stable optimization.

To further assess performance stability, Table III reports a quantitative comparison with three representative open-sourced SSCD methods under the challenging 5% labeled-data setting. SemiBCD consistently exhibits smaller standard deviations than all baseline methods, indicating more stable training behavior across multiple runs. In contrast, FixMatch and UniMatch achieve lower mean performance and show larger variability, particularly on the WHU-CD dataset, suggesting reduced robustness under extremely limited supervision. Overall, these results indicate that SemiBCD delivers more consistent and reliable improvements over the comparison semi-supervised approaches when only a small fraction of labeled data is available.

Fig. 7 provides a qualitative comparison of predictions under the 5% label regime. FixMatch and UniMatch often produce fragmented or noisy CDs, while SemiCD-VL struggles to accurately delineate building boundaries in dense urban areas. In contrast, SemiBCD preserves fine-grained structural details, yields sharp building outlines, and effectively suppresses false alarms in unchanged regions. Together, the quantitative and qualitative results demonstrate the effectiveness of our SemiBCD as a semi-supervised building CD framework.

C. Cross-Dataset Generalization

We further assessed the *cross-dataset generalization* of SemiBCD by training on a mixed set comprising labeled LEVIR-CD samples and unlabeled WHU-CD data, and testing on LEVIR-CD. This challenging setup evaluates the model's ability to transfer knowledge across datasets with distinct domain characteristics.

Table IV shows that SemiBCD consistently outperforms competing methods across all label ratios. Under the 5% label regime, SemiBCD achieves an IoU^c of 78.73% and an OA of 98.83%, surpassing SemiCD-VL by 5.56% in IoU^c. At 40% labeled data, it reaches an IoU^c of 80.79%, outperforming SemiCD-VL by 1.35%. Even with 100% labels, SemiBCD leads (IoU^c = 81.46% and OA = 99.00%), indicating its effectiveness in leveraging external unlabeled data.

These results highlight SemiBCD's strong generalization capability. By integrating VLM-guided pseudo-labeling with dual consistency learning, our framework shows promise in bridging domain gaps, suggesting its potential suitability for real-world applications where large-scale labeled datasets are impractical.

D. Inference Efficiency

We compared SemiBCD's inference speed with representative SSCD methods, as summarized in Table V. All models were tested under the same hardware environment to ensure fair comparison, and the frames-per-second (FPS) metric was used to assess runtime efficiency. Note that SemiBCD and SemiCD-VL share the same building CD backbone.

SemiBCD achieves an inference speed of 211.43 FPS, matching SemiCD-VL and surpassing all other baselines. In particular, it outperforms classical semi-supervised methods such as FixMatch and UniMatch (both at 191.23 FPS), and is substantially faster than earlier approaches including s4GAN (75.23 FPS) and SSCD (62.51 FPS). This demonstrates that SemiBCD delivers competitive performance while maintaining high inference efficiency.

E. Performance Comparisons With Supervised Methods

To further validate the effectiveness of our semi-supervised approach, we compared its performance with leading fully supervised CD methods on the LEVIR-CD and WHU-CD datasets, including MFPNet [52], BiT [14], ChangeFormer [55], AMTNet [17], and MCTNet [1]. The goal of this analysis is to determine whether our framework, with limited labeled data, can achieve performance comparable to or even better than models trained on a full set of pixel-level annotations. Following these supervised methods, we adopted F_1^c and IoU^c as evaluation metrics.

As shown in Table VI, even when trained with only a small fraction of the labeled data, our SemiBCD achieves performance that is highly competitive with, and in some cases surpasses, the best fully supervised models. On the LEVIR-CD dataset, SemiBCD with just 5% of the labeled data achieves an F_1^c score of 90.81% and an IoU^c of 83.17%. This is a remarkable result, as it is very close to the performance of fully supervised methods like MFPNet (F_1^c of 91.08%) and MCTNet (F_1^c of 90.98%), which were trained on 100% of the labeled data. Furthermore, SemiBCD with only 10% of the labels achieves the highest F_1^c (91.16%) and IoU^c (83.75%) on the LEVIR-CD dataset, outperforming all supervised baselines. This demonstrates that our method effectively leverages unlabeled data to bridge the performance gap between semi-supervised and fully supervised models.

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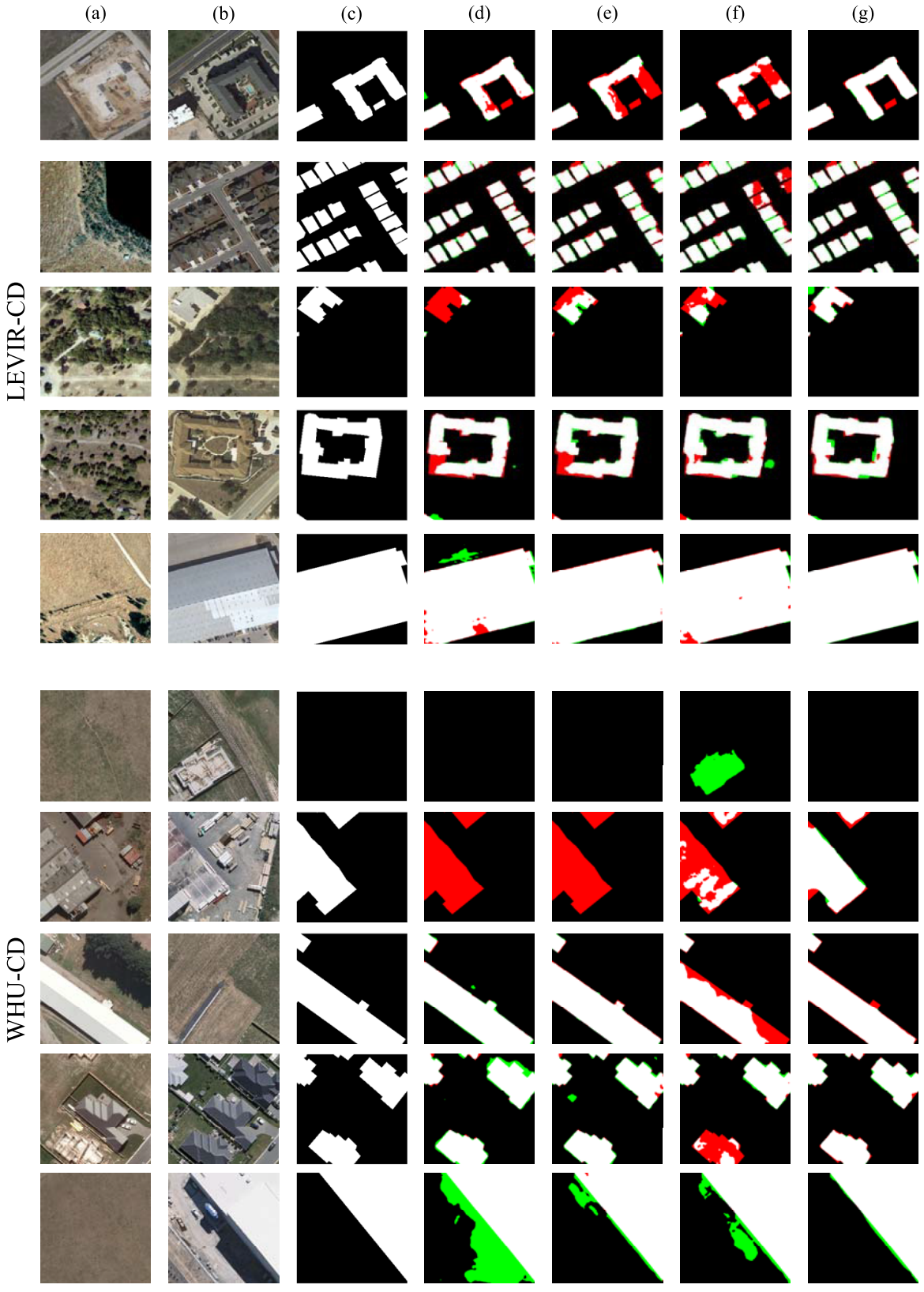


Fig. 7. Qualitative comparison of building CD results under the 5% labeled-data setting on LEVIR-CD and WHU-CD. (a) Pre-event image. (b) Postevent image. (c) Ground truth. (d) FixMatch. (e) UniMatch. (f) SemiCD-VL. (g) SemiBCD. False positives (FPs) are green, and false negatives (FNs) are red. A better result is indicated by fewer red and green pixels. For more details, please view in color and zoom in.

Similarly, on the WHU-CD dataset, SemiBCD with 5% labeled data achieves an F_1^c of 90.91% and an IoU^c of 83.34%. This result is better than that of BiT and is only slightly lower than that of the top-performing fully supervised methods. With

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TABLE IV

CROSS-DATASET GENERALIZATION RESULTS ON LEVIR-CD, TRAINED WITH VARYING PERCENTAGES OF LABELED LEVIR-CD DATA AND UNLABELED WHU-CD DATA

Method	{LEVIR(sup.%),WHU(unsup.)} → LEVIR				
	5% (IoU^c / OA)	10% (IoU^c / OA)	20% (IoU^c / OA)	40% (IoU^c / OA)	100% (IoU^c / OA)
AdvNet [48]	65.2 / 98.10	71.6 / 98.42	75.0 / 98.63	76.5 / 98.77	77.4 / 98.78
s4GAN [49]	66.7 / 98.12	68.4 / 98.26	73.4 / 98.54	75.3 / 98.65	75.8 / 98.63
SemiCDNet [6]	67.6 / 98.19	71.9 / 98.46	74.9 / 98.62	76.4 / 98.70	77.9 / 98.80
SemiCD [7]	71.4 / 98.40	74.6 / 98.58	76.4 / 98.69	77.9 / 98.77	78.6 / 98.82
SemiCD-VL [46]	73.17 / 98.55	77.84 / 98.81	78.38 / 98.85	79.44 / 98.87	80.69 / 98.95
SemiBCD	78.73 / 98.83	79.92 / 98.90	80.30 / 98.91	80.79 / 98.97	81.46 / 99.00

TABLE V

TESTING TIME EFFICIENCY COMPARISON

Method	Backbone	FPS
AdvNet	ResNet101	146.91
s4GAN	ResNet101	75.23
FixMatch	ResNet50	191.23
UniMatch	ResNet50	191.23
SemiCDNet	UNet++	151.70
SemiCD	ResNet50	62.51
SemiCD-VL	ResNet50	211.43
SemiBCD	ResNet50	211.43

TABLE VI

PERFORMANCE COMPARISONS WITH SUPERVISED METHODS ON THE LEVIR-CD AND WHU-CD DATASETS

Method	LEVIR-CD		WHU-CD	
	F_1^c	IoU^c	F_1^c	IoU^c
100% labeled data:				
MFPNet [52]	91.08	83.62	88.73	79.74
BiT [14]	89.31	80.68	85.83	75.18
ChangeFormer [55]	90.40	82.50	89.90	81.60
AMTNet [17]	90.76	83.08	92.27	85.64
MCTNet [1]	90.98	83.46	92.79	86.55
SemiCD-VL (5% labeled data)	90.01	81.83	89.90	81.65
SemiCD-VL (10% labeled data)	90.42	82.51	90.70	82.98
SemiBCD (5% labeled data)	90.81	83.17	90.91	83.34
SemiBCD (10% labeled data)	91.16	83.75	91.49	84.32

10% of the labeled data, SemiBCD secures the second-best performance on WHU-CD, with an F_1^c of 91.49% and an IoU^c of 84.32%, showing a similar performance to the best ones. These results indicate the superior performance of our SemiBCD.

F. Ablation Study

To assess the contribution of each key component in our framework, we conducted ablation experiments on the LEVIR-CD and WHU-CD datasets using 5% labeled data. Specifically, we analyzed the effectiveness of the UAPLR module and the impact of different loss functions.

1) Effectiveness of the UAPLR Module: We first examined the role of the UAPLR module. As shown in Table VII, removing the UAPLR module (denoted as "w/o") leads to a consistent but dataset-dependent performance degradation. On LEVIR-CD, the F_1^c and IoU^c decrease by 0.15% and 0.25%, respectively. The effect is more noticeable on WHU-CD, where there is a reduction of 1.34% in F_1^c and 2.23% in IoU^c. The impact of UAPLR appears to be more significant in complex urban environments such as WHU-CD, which feature diverse building layouts and greater challenges from pseudo-label noise. The qualitative comparisons in Fig. 8 further illustrate that the initial pseudo-labels provide moderate yet informative cues for CD. By applying UAPLR, these cues are effectively refined, resulting in more spatially coherent pseudo-labels and consequently cleaner and more accurate change mask predictions compared to the variant without refinement.

TABLE VII

ABLATION STUDIES OF THE UAPLR MODULE ON LEVIR-CD AND WHU-CD DATASETS (WITH 5% OF TRAINING DATA LABELED)

UAPLR	LEVIR-CD		WHU-CD	
	F_1^c	IoU^c	F_1^c	IoU^c
w/o	90.66	82.92	89.57	81.11
w/	90.81	83.17	90.91	83.34

TABLE VIII

ABLATION STUDY OF THE UAPLR MODULE ON BUILDING SEGMENTATION PERFORMANCE USING THE WHU BUILDING DATASET

UAPLR	IoU	Precision	Recall	F1
w/o	61.22	86.10	67.94	75.95
w/	**82.78**	**90.83**	**90.33**	**90.58**

To further quantify the effect of UAPLR on pseudo-label quality, we additionally conducted an experiment on the WHU Building dataset, as summarized in Table VIII. When UAPLR is disabled, the VLM already achieves reasonable building segmentation performance, attaining an IoU of 61.22% and an $F1$-score of 75.95%.

After enabling UAPLR, consistent and substantial improvements are observed across all evaluation metrics, building upon the initially high-quality pseudo-labels. Specifically, the IoU increases from 61.22% to 82.78%, while recall and $F1$ improve from 67.94% to 90.33% and from 75.95% to 90.58%, respectively. These results indicate that UAPLR effectively filters unreliable pseudo-label regions and improves the overall quality of pseudo-labels, which contributes to more stable training and better downstream performance.

2) Impact of Different Loss Functions: We further analyzed the effect of each loss term in the overall objective function.

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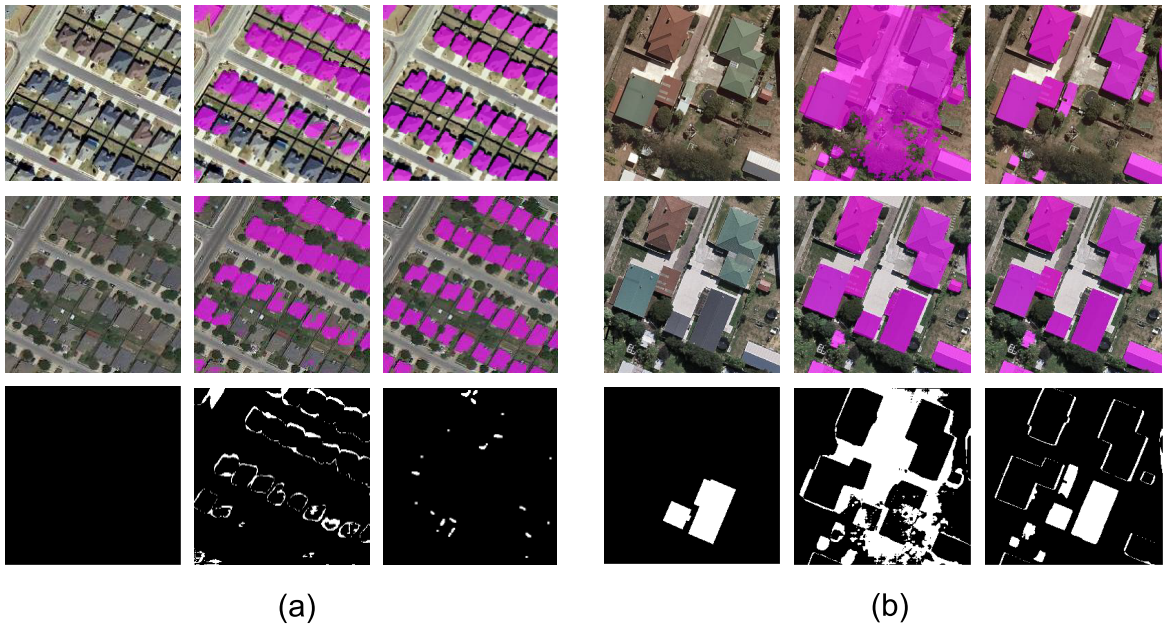


Fig. 8. Qualitative results illustrating the effect of the UAPLR module on (a) LEVIR-CD and (b) WHU-CD. Each group corresponds to a bitemporal image pair: the first column shows the pre- and postevent images along with the ground-truth change mask. Columns 2 and 3 display the building pseudo-labels generated without and with the UAPLR module, respectively.

TABLE IX

ABLATION STUDIES OF THE LOSS FUNCTIONS ON LEVIR-CD AND WHU-CD DATASETS (WITH 5% OF TRAINING DATA LABELED)

Method	LEVIR-CD		WHU-CD	
	F_1^c	IoU^c	F_1^c	IoU^c
$\mathcal{L}_s$	89.29	80.66	86.09	75.57
$\mathcal{L}_s + \mathcal{L}_{mvc}$	90.67	82.94	90.15	82.06
$\mathcal{L}_s + \mathcal{L}_{tha}$	89.80	81.49	88.30	79.04
SemiBCD	90.81	83.17	90.91	83.34

TABLE X

PERFORMANCE UNDER DIFFERENT VALUES OF λ_1 AND λ_2 (WITH 5% OF TRAINING DATA LABELED)

λ_1	λ_2	LEVIR-CD		WHU-CD	
		F_1^c	IoU^c	F_1^c	IoU^c
0.0	0.0	89.29	80.66	86.09	75.57
0.1	0.1	89.52	81.03	90.57	82.77
0.2	0.2	90.83	83.21	90.88	83.28
0.3	0.3	90.81	83.17	90.91	83.34
0.4	0.4	90.71	83.00	90.99	83.47
0.5	0.5	90.85	83.23	90.81	83.17
0.6	0.6	90.55	82.73	90.68	82.95
0.7	0.7	90.61	82.84	90.78	83.12
0.3	0.0	90.67	82.94	90.15	82.06
0.0	0.3	89.80	81.49	88.30	79.04

The supervised loss ($\mathcal{L}_s$) was used as the baseline, and additional terms were incrementally introduced.

As shown in Table IX, semi-supervised objectives consistently outperform the supervised-only baseline. Incorporating the multiview consistency loss ($\mathcal{L}_{mvc}$) yields substantial improvements: the F_1^c increases from 89.29% to 90.67% on LEVIR-CD and from 86.09% to 90.15% on WHU-CD. This underscores the importance of consistency regularization in exploiting unlabeled data for improved robustness and generalization.

Adding the temporal consistency loss ($\mathcal{L}_{tha}$) also enhances performance compared to the baseline although the gains are smaller than those provided by $\mathcal{L}_{mvc}$. With $\mathcal{L}_s + \mathcal{L}_{tha}$, the model achieves F_1^c scores of 89.80% on LEVIR-CD and 88.30% on WHU-CD.

The full SemiBCD framework, which integrates all three losses ($\mathcal{L}_s + \mathcal{L}_{mvc} + \mathcal{L}_{tha}$), achieves the best performance on both datasets. On LEVIR-CD, it reaches an F_1^c of 90.81% and IoU^c of 83.17%, slightly surpassing the $\mathcal{L}_s + \mathcal{L}_{mvc}$ variant. On WHU-CD, the full model shows a larger margin, outperforming the $\mathcal{L}_s + \mathcal{L}_{mvc}$ variant by 0.76% in F_1^c. These results confirm that each loss term contributes positively, and their combination provides complementary benefits.

G. Parameter Sensitivity Analysis

We conducted a sensitivity study to evaluate the robustness of SemiBCD with respect to three key hyperparameters: the consistency regularization weights λ_1 and λ_2, as well as the uncertainty threshold τ_{low} used in UAPLR. All experiments were performed under the challenging 5% labeled-data regime to highlight model behavior when supervision is scarce.

1) Loss Weights λ_1 and λ_2: To streamline the experimental setup and reduce the considerable workload of hyperparameter tuning, we analyzed a set of symmetric configurations where $\lambda_1 = \lambda_2$. While setting these two complementary terms to be equal may not yield the absolute mathematical optimum, this choice provides a highly effective and robust configuration, as demonstrated by the consistently high performance achieved

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TABLE XI
PERFORMANCE UNDER DIFFERENT VALUES OF τ_{LOW} (5% OF TRAINING DATA LABELED)

τ_{low}	LEVIR-CD		WHU-CD	
	F_1^c	IoU^c	F_1^c	IoU^c
0.1	90.52	82.68	88.87	79.97
0.15	90.59	82.80	89.56	81.10
0.2	90.81	83.17	90.91	83.34
0.25	90.89	83.31	89.65	81.25
0.3	90.82	83.18	88.45	79.30

when $\lambda_1 = \lambda_2$ in Table X. This analysis reveals that both MVCR and THA are essential and complementary; setting both to zero reduces SemiBCD to a purely supervised baseline, yielding the lowest performance. Gradually increasing the weights from 0 to 0.3 consistently improves accuracy, with peak results typically observed at $\lambda_1 = \lambda_2 = 0.3$. Performance becomes slightly unstable when either weight exceeds 0.6, suggesting that excessive regularization can hinder the learning process. Notably, testing asymmetric configurations leads to significant drops in performance, confirming that the synergy between the two consistency mechanisms is critical for achieving strong results.

2) Uncertainty Threshold τ_{Low}: Table XI further demonstrates the effectiveness of UAPLR. Very low thresholds (e.g., 0.1) retain noisy pseudo-labels, degrading performance—especially on the more complex WHU-CD dataset. The best and most stable results are achieved with moderate thresholds between 0.2 and 0.25, which balance label retention and noise suppression. Performance declines when $\tau_{\text{low}} > 0.3$, indicating that over-filtering discards useful supervisory signals. Overall, SemiBCD maintains consistent performance across multiple tested hyperparameter configurations, especially with moderate consistency weights and $\tau_{\text{low}} = 0.2$.

V. CONCLUSION

This article presented *SemiBCD*, a semi-supervised framework for building CD that integrates VLM priors with consistency learning. By combining VLM-based pseudo-label generation, UAPLR, and dual consistency regularization, SemiBCD aims to reduce reliance on dense annotations while maintaining competitive performance under limited supervision. Experiments on the LEVIR-CD and WHU-CD benchmarks, together with supplementary analysis on the WHU Building dataset, demonstrate that the proposed method achieves consistent improvements over representative semi-supervised baselines in low-label settings.

Several limitations should be acknowledged. The experimental evaluation is restricted to building CD in urban and suburban environments, and the framework has not yet been validated on nonbuilding or multiclass CD tasks. In addition, the method relies on pseudo-labels generated by a pretrained VLM, whose effectiveness may be affected by domain shifts between natural images and remote sensing imagery. Although uncertainty-aware refinement alleviates some noise, it does not fully eliminate this dependence. Moreover, the framework introduces multiple hyperparameters and additional training-time computational overhead due to VLM fine-tuning and pseudo-label refinement, which were not included in the reported inference-time efficiency analysis.

Future work will explore extending the framework to more diverse CD scenarios, investigating adaptive hyperparameter selection strategies, and evaluating alternative or domain-adapted VLM backbones.

REFERENCES

[1] W. Liu, Z. Kang, J. Liu, Y. Lin, Y. Yu, and J. Li, "A multitask CNN-transformer network for semantic change detection from bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5647215.

[2] A. Tayfehrostami, A. Abedini, and B. Tajfirooz, "Analyzing the variations in the water surface area of taleqan dam of Iran using ground-based and satellite observations," in *Proc. Int. Conf. Mach. Intell. Geoanal. Remote Sens. (MIGARS)*, vol. 1, Jan. 2023, pp. 1–4.

[3] H. Sabeti, A. Pourmina, A. Rezaei, A. Tayfehrostami, F. Einlou, and H. Nankali, "Discovering confined zones and land deformation characteristics across an aquifer system in Iran using GNSS and InSAR techniques," *Hydrogeol. J.*, vol. 31, no. 8, pp. 2061–2076, 2023.

[4] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, and P. He, "SCDNET: A novel convolutional network for semantic change detection in high resolution optical remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102465.

[5] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[6] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[7] W. Gedara Chaminda Bandara and V. M. Patel, "Revisiting consistency regularization for semi-supervised change detection in remote sensing images," 2022, *arXiv:2204.08454*.

[8] C. Sun, H. Chen, C. Du, and N. Jing, "SemiBuildingChange: A semi-supervised high-resolution remote sensing image building change detection method with a pseudo bitemporal data generator," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5622319.

[9] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[10] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[11] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[12] W. Wang, X. Tan, P. Zhang, and X. Wang, "A CBAM based multiscale transformer fusion approach for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6817–6825, 2022.

[13] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[14] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[15] X. Song, Z. Hua, and J. Li, "PSTNet: Progressive sampling transformer network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8442–8455, 2022.

[16] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111.

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[17] W. Liu, Y. Lin, W. Liu, Y. Yu, and J. Li, "An attention-based multiscale transformer network for remote sensing image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 599–609, Aug. 2023.
[18] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4301, 2022.
[19] Y. Shen et al., "Aligning and prompting everything all at once for universal visual perception," in *Proc. CVPR*, Jun. 2024, pp. 13193–13203.
[20] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.
[21] H. Xia, Y. Tian, L. Zhang, and S. Li, "A deep Siamese postclassification fusion network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622716.
[22] Y. Niu, H. Guo, J. Lu, L. Ding, and D. Yu, "SMNet: Symmetric multi-task network for semantic change detection in remote sensing images based on CNN and transformer," *Remote Sens.*, vol. 15, no. 4, p. 949, Feb. 2023.
[23] F. Cui and J. Jiang, "MTSCD-Net: A network based on multi-task learning for semantic change detection of bitemporal remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103294.
[24] W. Zhang, X. Lu, and X. Li, "A coarse-to-fine semi-supervised change detection for multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3587–3599, Jun. 2018.
[25] J.-X. Wang, T. Li, S.-B. Chen, J. Tang, B. Luo, and R. C. Wilson, "Reliable contrastive learning for semi-supervised change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4416413.
[26] C. Sun, J. Wu, H. Chen, and C. Du, "SemiSANet: A semi-supervised high-resolution remote sensing image change detection model using Siamese networks with graph attention," *Remote Sens.*, vol. 14, no. 12, p. 2801, Jun. 2022.
[27] Q. Shu, J. Pan, Z. Zhang, and M. Wang, "MTCNet: Multitask consistency network with single temporal supervision for semi-supervised building change detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 115, Dec. 2022, Art. no. 103110.
[28] L. Wang, M. Zhang, and W. Shi, "STCRNet: A semi-supervised network based on self-training and consistency regularization for change detection in VHR remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2272–2282, 2024.
[29] Z. Mao, X. Tong, and Z. Luo, "Semi-supervised remote sensing image change detection using mean teacher model for constructing pseudo-labels," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.
[30] L. Wang, J. Zhang, D. Chen, and L. Bruzzone, "PAdaptCD: Progressive adaptive thresholding and bitemporal image augmentation for semisupervised change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 17687–17705, 2025.
[31] W. Zhang, X. Shu, S. Wu, and S. Ding, "Semi-supervised change detection with data augmentation and adaptive thresholding for high-resolution remote sensing images," *Remote Sens.*, vol. 17, no. 2, p. 178, Jan. 2025.
[32] C. Wu, B. Du, and L. Zhang, "Fully convolutional change detection framework with generative adversarial network for unsupervised, weakly supervised and regional supervised change detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9774–9788, Aug. 2023.
[33] Y. Wang, J. Sha, L. Gao, Y. Zhang, X. Rong, and C. Zhang, "A semi-supervised domain alignment transformer for hyperspectral images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525411.
[34] K. Ahn, S. Han, S. Park, J. Kim, S. Park, and M. Cha, "Generalizable disaster damage assessment via change detection with vision foundation model," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2025, vol. 39, no. 27, pp. 27784–27792.
[35] R. Fan, J. Xie, J. Liu, Y. Zhang, H. Hou, and J. Yang, "DPCA: Dynamic multi-prototype cross-attention for change detection unsupervised domain adaptation of remote sensing images," *Knowl.-Based Syst.*, vol. 314, Apr. 2025, Art. no. 113135.
[36] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.
[37] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.
[38] Y. Yang, X. Tang, J. Ma, X. Zhang, S. Pei, and L. Jiao, "ECPS: Cross pseudo supervision based on ensemble learning for semi-supervised remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612317.
[39] S. Yuan, R. Zhong, C. Yang, Q. Li, and Y. Dong, "Dynamically updated semi-supervised change detection network combining cross-supervision and screening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5613514.
[40] Z.-H. You, S.-B. Chen, J.-X. Wang, C. H. Q. Ding, J. Tang, and B. Luo, "Prototype discriminative learning for semi-supervised change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4416114.
[41] J. Shen, C. Zhang, M. Zhang, Q. Li, and Q. Wang, "Learning remote sensing aleatoric uncertainty for semi-supervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635413.
[42] L. Ran, D. Wen, T. Zhuo, S. Zhang, X. Zhang, and Y. Zhang, "AdaSemiCD: An adaptive semi-supervised change detection method based on pseudo-label evaluation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5615814.
[43] D. Peng, M. Liu, and H. Guan, "Perturbation matters: A novel approach for semi-supervised remote sensing imagery change detection," *Remote Sens.*, vol. 17, no. 4, p. 576, Feb. 2025.
[44] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[45] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. Jorge Cardoso, "Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Proc. MICCAI*, 2017, pp. 240–248.
[46] K. Li et al., "SemiCD-VL: Visual-language model guidance makes better semi-supervised change detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5601613.
[47] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[48] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Pérez, "Advent: Adversarial entropy minimization for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 2517–2526.
[49] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2021.
[50] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.
[51] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.
[52] J. Xu, C. Luo, X. Chen, S. Wei, and Y. Luo, "Remote sensing change detection based on multidirectional adaptive feature fusion and perceptual similarity," *Remote Sens.*, vol. 13, no. 15, p. 3053, Aug. 2021.
[53] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
[54] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[55] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.



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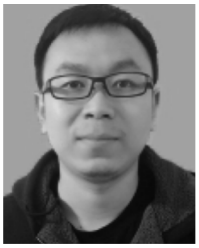
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